



The effect of predictive analytics-driven interventions on healthcare utilization

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ABSTRACT

This paper studies a commercial insurer-driven intervention to improve resource allocation. The insurer developed a claims-based algorithm to derive a member-level healthcare utilization risk score. Members with the highest scores were contacted by a care management team tasked with closing gaps in care. The number of members outreached was dictated by resource availability and not by severity, creating a set of arbitrary cutoff points, separating treated and untreated members with very similar predicted risk scores. Using a regression discontinuity approach, we find evidence that predictive analytics-driven interventions directed at high-risk individuals reduced emergency room and specialist visits, yet not hospitalizations.

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Introduction

The distribution of healthcare spending in the U.S. is highly skewed, with 21% of spending concentrated among the top 1% of spenders and nearly 50% of spending concentrated among the top 5% of spenders (Kaiser Family Foundation, 2010). A large body of literature shows that Medicare recipients consume a disproportionate share of health care services (Card, et al. 2009; Finkelstein, 2007; Niska et al., 2010). While people over age 64 in 2009 comprised 13% of the US population, they made up 38.2% of the top 5% and 40% of the top 1% of highest spenders (NIHCM Foundation Data Brief, 2012). Even among Medicare recipients, the driver of hospitalizations and ED visits are patients with chronic conditions such as congestive heart failure (AHRQ, 2013). To reduce unnecessary and preventable care, the fast-changing healthcare landscape is often characterized by a shift from fee-for-service with an emphasis on volume to a value-based reimbursement structure (Miller, 2009). To thrive in this new environment, providers must be proactive as opposed to reactive (i.e. anticipate problems

before they arise versus be there to escalate care when an individual patient's condition deteriorates). Similarly, insurance providers are interested in reducing preventable hospitalizations and shifting expensive emergency department (ED) visits to the primary care setting. One such proactive clinical decision-making approach entails the use of real time risk stratification of their patient population to close gaps in care and therefore prevent costly and potentially unnecessary service use (Bates et al., 2014).

The effort to contain cost, improve care quality and reduce wasteful care, by improving access to primary and preventive care has received attention in the literature. Cost and downstream healthcare utilization were influenced by a focus on population health management and continuity of care (Dusheiko et al., 2011; Kranker, 2016) as well as improved access to care (Alexander et al., 2017; Dobkin et al., 2016; Finkelstein et al., 2012; Taubman et al., 2014; Miller, 2012). Opportunities for closing gaps in care through enhanced coordination are likely large among the Medicare population. The typical Medicare beneficiary receives care from eight distinct providers per year while the top decile has seen over 21 providers per year (Agha et al., 2017). Care fragmentation is especially damaging for chronically ill patients and have led to adverse health outcomes (McWilliams, 2016; Milstein and Gibertson, 2009).

Most individuals, even those at high risk, are not medically supervised around the clock. Therefore, the ability to maintain an

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active lifestyle, balanced diet, medication adherence, and escalate care in the face of sudden health changes is essential for successful disease management. In a way, if the member is not very efficient in interpreting health signals and turning them into health actions, preemptive interventions by third party payers with access to detailed individual level data can strengthen the connection between observing certain health signals and effectively acting on them. Put differently, data-driven claims-based analysis can serve to raise the probability of healthy, cost-lowering, and quality enhancing choices for patients.

To our knowledge, this is the first evaluation of an intervention designed to improve care coordination and close gaps in care in a high-risk chronic population that was driven by a preemptive, data-driven, claims-based analysis. We utilize data from an insurer-driven intervention to reduce hospital and emergency department utilization among their fully-insured Medicare Advantage member population who were diagnosed with Congestive Heart Failure (CHF). Efforts to reduce healthcare utilization for Medicare CHF patients has the potential of leading to significant savings for the insurer and increase well-being for their members. The annual cost of avoidable or preventable hospitalizations and emergency room visits surpasses 25 billion dollars per year (AHRQ, 2004). There are more than 1 million CHF hospitalizations annually, with over 70% of hospitalizations in the Medicare population (NCHS Data Brief, 2012), and CHF accounted for 21.1% of all potentially avoidable hospitalizations in 2009 (Segal et al., 2014).

The insurer utilized a proprietary algorithm to derive a risk score for each member based on patterns of healthcare consumption, medication use, demographic characteristics, and additional proprietary member experience data from different sources. The higher the risk score, the more likely the member is to experience gaps in care and be at risk for hospitalization. A designated care coordination management team was tasked with identifying and closing gaps in care by contacting members with relatively high-risk scores. However, members were not chosen for treatment based on an absolute level of risk, but rather the care management team's resource availability determined how many members they could reach out to. Every time the algorithm was run it produced a ranking of members and a corresponding set of arbitrary cutoff points for each outreach wave, separating treated and untreated members with very similar predicted risk scores.

This study has two primary aims. First, we assess whether the algorithm-derived risk score is predictive of hospitalizations and ED visits. Second, we use an intent-to-treat approach to estimate the causal effect of the care management intervention on inpatient and outpatient healthcare utilization over time. Here, our main analyses utilize the ad-hoc cutoffs across waves and between treatment and control groups, and rely on a Regression Discontinuity (RD) design that compares individuals above and below the threshold of selection into the outreach program.

We find that the risk score is a strong predictor of hospitalization and ED visits 30, 90, 180 and 360 days after the algorithm was run. Our estimates suggest that the outreach program reduces the number of ED visits across all post treatment periods, and some evidence of a reduction in cardiologist visits. Specifically, we find that the number of outpatient ED visits decreases by about 40% within the first year following treatment. At the same time, while directionally similar, we find little or no statistically significant impact of the outreach program on the number of PCP visits or hospitalizations. Along the extensive margin, we find strong evidence that the probability of having any cardiologist and ED visit decreases sharply. Specifically, we find that in the short- and long-term the probability of having a cardiologist visit decreases by 14 percentage points. We also find stronger effects in the long-run for ED, suggesting that the probability of having an ED visit decreases by

12 percentage points. In terms of PCP visits and hospitalizations, we find no changes in the probability of having such event. We conduct several robustness checks to test the validity of the RD design and sample selection. In all cases, our results are relatively stable and unaffected by the change of specification and bandwidth selection.

Background

While the use of predictive analytics to manage population health has been widely discussed, there is little research to suggest that predictive analytics-driven interventions are successful. Several randomized trials have evaluated the impact of care management programs, in some cases among populations with chronic illnesses, and for the most part find little or no improvements in clinical outcomes or spending (Bleich et al., 2015). The reason for the weak findings may be the result of broad inclusion criteria leading to a combination of people who stand to benefit from the intervention and people who do not. These findings may differ under predictive analytics-driven targeting of at-risk populations, as in the case of our study.

Existing evidence on the benefits of care continuity,¹ especially in primary care, for the elderly is also typically not driven by predictive analytics, but has shown to reduce ED visits and hospitalizations (Nyweide and Bynum, 2017; Amjad et al., 2016 and Hussey et al., 2014). Concerns regarding these papers' findings remain, as members that are in better unobservable physical and mental health are more likely to experience continuous care and less likely to require hospitalizations and ED visits. In this case the relationship between measures of care continuity and utilization are driven by unobserved propensity to use care, which is addressed in our study by predictive analytics.

Continuity of care does not automatically translate into reductions in unnecessary care among the general population. Three recent systematic reviews of randomized controlled trials find limited health benefits of having an annual check-up with a primary care physician (PCP) on hospitalizations and mortality (Krogsbøll et al., 2012; Boulware et al., 2007; Peikes et al., 2009). Similarly, quasi experimental evidence among patients who consistently received an annual physician check-up shows that a notice of elevated levels of blood sugar (leading to a pre-diabetes diagnosis) causes additional preventive care, but suggests that such increase in care utilization does not translate into health improvements (Iizuka et al., 2017). On the other end, Bruni et al. (2016) and Dolton and Pathania (2016) show that increased primary care hours were associated with a reduction in ED visits by about 10%. Similarly, the adoption of the patient centered medical home model, design to improve accessibility and continuity of care was associated with 5–8% reduction in ED use (David et al., 2015).

Data

Our data were obtained from Independence Blue Cross of Philadelphia and consist of all insurance claims for 29,222 fully insured Medicare Advantage members residing in the Philadelphia area who were diagnosed with CHF. The model is a proprietary predictive model developed by the health plan using SAS Enterprise Miner. The model uses over 500 patient-level features derived from medical and pharmacy claims, lab results, customer service contacts, referral authorizations, clinical alerts, demographic information, and features derived from natural language processing

¹ Increased provider continuity is typically measured as the share of healthcare visits for a unique provider or the total number of visits within a certain time frame.

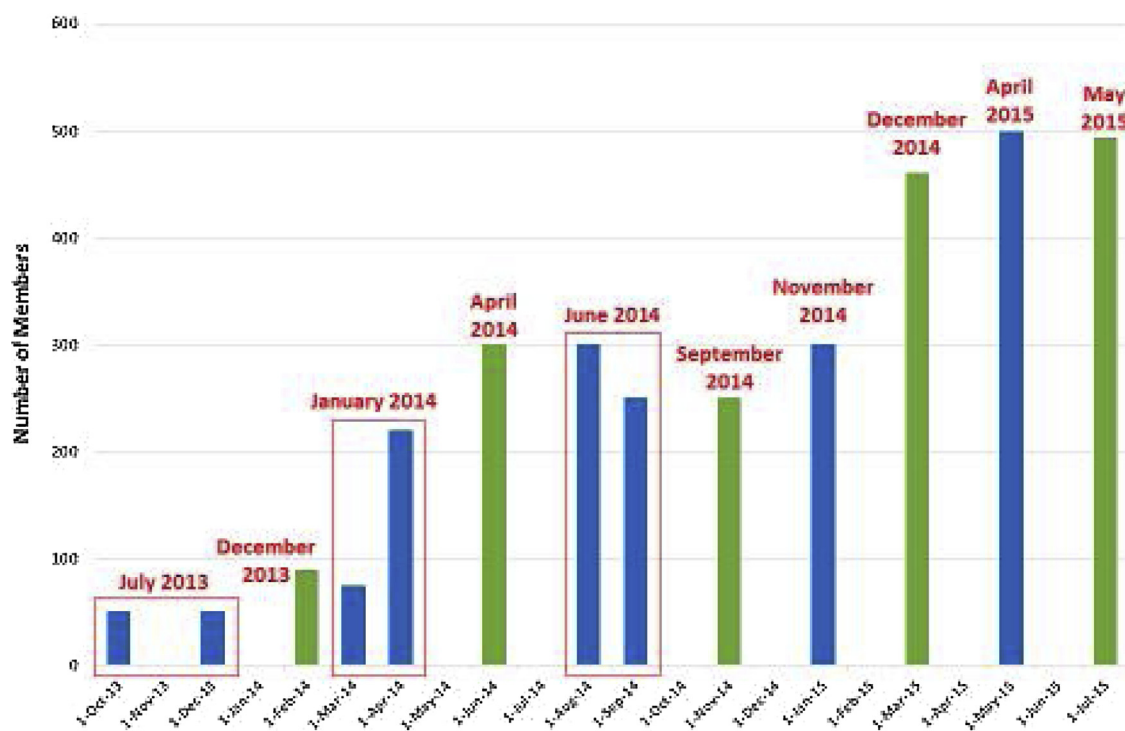


Fig. 1. Dates of Outreach by Wave and Number of Members Outreached.

Notes: Fig. 1 outlines the date the member information was forwarded to the care management team (x-axis), the number of members sent to the care management team (y-axis), and the date the algorithm was run (appears above the bars).

analysis of comments and customer call notes, to predict patient-level risk of hospitalization and ED visits.²

The predictive algorithm was applied in 10 waves between July 2013 and May 2015. Each wave included the assignment of an individual risk score to all members who met the inclusion criteria above. Members were ranked from highest to lowest on the predicted risk score. At each wave the care management (CM) team determined how many members they could outreach to and requested from the Independence data team a list of between 50 and 500 members, starting with the highest risk. Fig. 1 outlines the date the member information was forwarded to the CM team (x-axis), the number of members sent to the CM team (y-axis), and the date the algorithm was run (appears above the bars). Overall, the later waves produced larger outreach cohorts than the earlier waves. The time between algorithm execution and transmission date to the CM team averaged about 2 months. The information for the early waves 1, 3 and 5 were forwarded in two batches to the CM team while the information of all other waves was forwarded at in a single date.³ For example, the first algorithm ran in July 2013 and 100 members were identified to have their information forwarded. Information for 50 members with the highest risk scores was sent to the CM team on 10/7/2013 and the information for the remaining 50 members with the next highest risk scores was sent on 12/2/2013. The algorithm ran for the second time in December 2013 and provided the basis for an information transmission on 3/19/2014 for 89 members in one batch. Since the program used sampling with

replacement across waves, our analyses exclude members included in subsequent outreach cohorts (i.e. we make sure that a member, whether in treatment or control groups, does not appear in more than one wave).

The quasi-random assignment into the program occurred because at each point the CM team determined how many members they could outreach to based on resource availability. This created a set of arbitrary cutoff points for each wave, separating treated and untreated members with very similar predicted risk scores and other demographic characteristics. The waves vary in time and size, and therefore produced different ranges of risk scores as well as different cutoff points. This heterogeneity is shown in Fig. 2, where for each wave, numbered on the x-axis, we plot the maximum risk score (i.e. the highest risk-score value in the treatment group), the value of the risk-score at the cutoff point, and the minimum risk score (i.e. the lowest risk-score value in the control group). Within wave threshold risk-scores vary between 0.462 and 0.616. Waves 1 through 3 have the lowest risk scores at cutoff, waves 4 through 7 have the highest risk scores at cutoff, and waves 8 through 10 have risk score cutoffs that lie in-between these values. This raises the importance of using *within*-wave variation as well as providing some wave specific analysis to capture algorithm improvements and changes in the underlying patient population over time.

Treatment

Telephone outreach was delivered by registered nurse health coaches employed by Independence Blue Cross. For each case, the health coach conducts a needs assessment to identify the specific individual-level factors that are known or believed to be associated with the members' likelihood of being hospitalized in the next 6 months. The health coach then works with the member to address any issues that may contribute to the likelihood of being hospitalized. In addition to helping members understand their diagnosis and how to manage it, health coaches often play a larger role in

² The sample size displays all members observed at some point during our sample period. We focus our analysis on patients who were most similar to our treatment group across all waves in terms of the observed risk score. We choose to focus on patients with similar risk scores around the treatment cutoff point to limit unobservable factors influencing our estimates. This restriction limits our sample given that we have only about 3,500 treated patients across all waves.

³ The selection of sending early waves in batches is due to administrative constraints and due to other priorities of the care management team at the time.

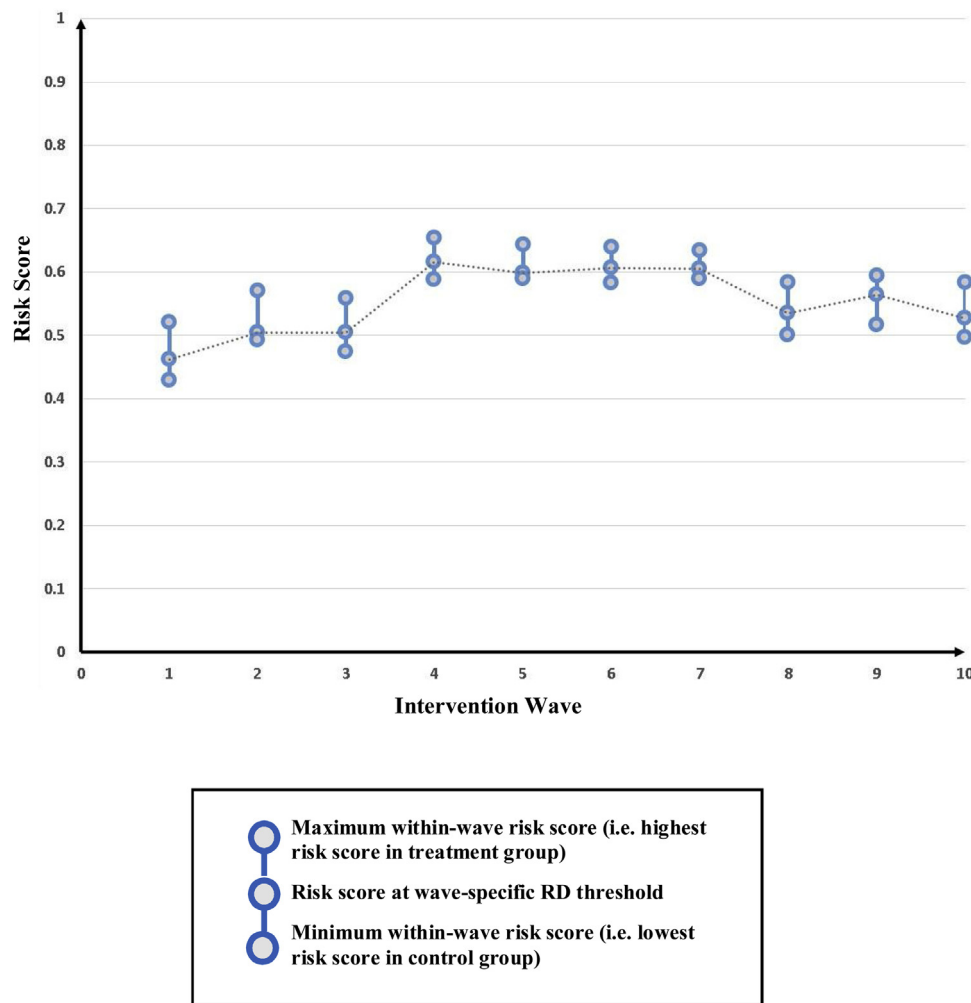


Fig. 2. Variation in Risk Score Range and RD Cutoff Point across Intervention Waves.

Notes: Fig. 2 displays the minimum, maximum and risk score at the cutoff threshold for each wave.

coordinating care by talking directly to the member's physicians, coordinating appointments, assisting with access to medications and medication adherence, arranging transportation, and involving social workers or behavioral health specialists when needed. In a way, the intervention can be viewed as a nudge for members to seek potentially necessary care.

Our analysis focuses on comparing members selected for outreach because we do not have information regarding the actual enrollment into the care management program,⁴ which constitute an intent to treat (ITT) group, with members with similar risk scores who were not selected for outreach. Outreach was initiated through phone calls, but not all calls resulted in enrollment in a care management program. Some calls were not answered and in other cases, members received the call but declined the intervention.

Moreover, the CM outreach alone might be considered a mild intervention, potentially prompting members to evaluate their health status and may prompt behavior change such as consulting a doctor or adhering to medication. Note that all members in this study had an assigned PCP and had utilized care that produced data, on which the predictive algorithm was run.

Mechanisms

As a result of the outreach there are many potential channels that could affect healthcare utilization. First, simply informing members that they were found to be at high risk of health deterioration may serve as a trigger for behavior change, such as better diet, improved medication adherence, and stronger reliance on primary care (though increased primary care reliance has rarely been found in previous studies). Second, enrollment in a care management program has the potential to have the member better integrated to the health care system, which may result in early detection of potential health deteriorations and reduce the reliance on traditional face-to-face interaction with primary care providers. Third, care managers serve as a communication hub who provided health education and comprehensive medication management. The latter two aspects have been shown to reduce care utilization across the board for chronically ill members, suggesting that one can also expect to find similar results in this intervention.

⁴ Based on anecdotal evidence about 50% of members enrolled in some sort of care management program that varied in intensity from pure informative and condition management to more directly condition specific therapy. The likely impact of the intervention is twofold, first, the outreach itself can be seen as a form of intervention by updating the members beliefs about their current health status and the potential risk if no action is taken. Second, the outreach provides resources to improve chronic disease management with an emphasis on preventing future negative health events and coordinates existing care for members who may sometimes lack the ability to schedule care in an appropriate time period to avoid the potential for emergency or unplanned care needs.

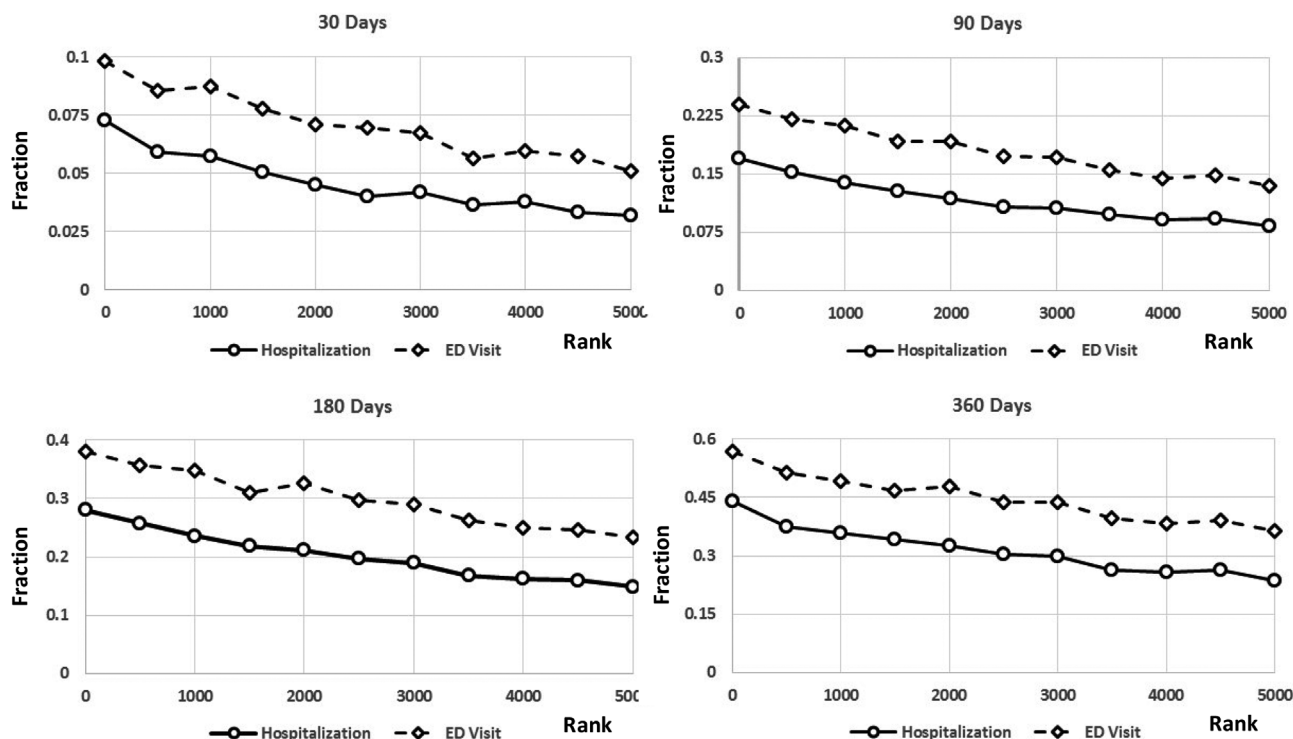


Fig. 3. Fraction of Hospitalizations and ED Visits for Members not Selected for Outreach by Post-Outreach Windows and Risk Score Rank (with 1 being highest risk score and 5000 being the lowest risk score).

Notes: Fig. 3 displays the probability of having a hospital admission or ED visit for all members who did not receive the outreach rank ordered based on their risk score from 0 to 5000 (with 0 being the member closest to the cutoff point).

Study design

We begin by evaluating whether the risk score was effective in predicting healthcare utilization. This tells us whether or not the algorithm is effective in targeting members at elevated risk of health deterioration. Fig. 3 shows the fraction of actual hospitalizations and ED visits within 30, 90, 180 and 360 days from the time of outreach initiation in a sample of all individuals who were not selected for outreach, as they were below the threshold (control group).⁵ As expected, and regardless of the follow up time window, the number of hospitalizations decreases as the relative distance from the cutoff increases. For example, 10% of members at the cutoff point had at least one ED visit within 30 days, while the frequency for members ranked around the 5000 mark was only 5%.⁶ These frequencies are higher for ED visits relative to hospitalizations and, of course, increase with the size of the follow-up window.⁷

To accomplish the second study aim, we focus on all individuals who were selected for treatment and form a similar sized control group consisting of individuals whose risk score rankings fall just below the treatment cutoff (threshold). We therefore select the control group by continuously selecting candidates along the risk score who would be the next in line to receive the intervention. We dropped individuals who were selected for outreach a second time.

⁵ Member attrition over our sample period is low. Conditional on being scored in a wave, the likelihood of being scored in a subsequent wave was 98%. While we cannot verify the cause of attrition (e.g. switching to a different insurance company, moving to a different state, or death), attrition is balanced across treatment and control groups in our sample.

⁶ Notice that there is a large negative correlation (-0.7) between rank and risk score. The higher the risk score the lower the rank.

⁷ The predictive power of the algorithms was tested for the goals set by the program, i.e. reductions in hospitalizations and ED visits. Similar trends, maybe not surprisingly, were not found for primary and specialist visits.

In the full sample, the ITT group has significantly higher risk scores than the control group by design, mostly due to unusually high risk score for those at the very top of the list. In the context of a regression discontinuity (RD) setting, we narrowed the bandwidth around the threshold, which resulted in reduced bias at the expense of increased variance (lower statistical power). We apply a bandwidth of 125, given the homogeneity of the sample (Table 1), but explored different bandwidth sizes in our sensitivity analyses. We provide estimates of alternative bandwidths in the robustness section.⁸ This allows for similar risk scores across the treatment and control groups and a large enough sample to detect a treatment effect across all 10 waves. After these sample modifications, we are left with a sample of 1974 individuals of which 987 are in the treatment and 987 are in the control group. In addition, our data include information on member's age, gender, insurance product type (HMO vs. PPO), whether the individual's insurance coverage includes a pharmacy benefit, and the length of enrollment (in months) in a Medicare Advantage plan. Finally, we use the claims data to identify all dates of primary care visits, cardiologist visits, emergency room visits, and hospitalizations.

These data allow us to evaluate whether a person received inpatient or outpatient care, and the timing of care relative to the timing

⁸ The bandwidth selection was performed based on comparability of the sample in terms of observable characteristics and sharp regression discontinuity estimates based on the methods provided by Calonico et al. (2014) and Imbens and Kalyanaraman, (2012). The sharp regression discontinuity procedures consistently suggested a bandwidth fluctuating around 100–150 (sometimes more sometimes less depending on the outcome variable). However, their methods do not allow for the inclusion of control variables, and our model can therefore absorb additional variation to estimate more precise effects and also control for bias even in cases when we choose a larger bandwidth.

Table 1
Summary Statistics for the Sample Stratified by Treatment and Control Group.

	Intent to Treat	Control
<i>Demographic controls</i>		
Risk Score ^a	0.61 (0.23)	0.50 (0.21)
Age	80.42 (8.25)	79.59 (8.56)
Enrolled Months	29.97 (6.32)	29.96 (5.77)
Female	0.56 (0.50)	0.56 (0.50)
HMO	0.94 (0.24)	0.95 (0.22)
Pharmacy Coverage	0.99 (0.09)	0.99 (0.09)
<i>Continuous Dependent Variables</i>		
	Intent to Treat	Control
PCP 30 days	0.47 (1.14)	0.36 (1.03)
PCP 90 days	1.18 (2.38)	0.97 (2.18)
PCP 180 days	2.30 (4.09)	1.86 (3.46)
PCP 360 days	4.01 (5.94)	3.42 (5.49)
Cardio 30 days	0.25 (0.55)	0.25 (0.55)
Cardio 90 days	0.75 (1.13)	0.73 (1.12)
Cardio 180 days	1.40 (1.82)	1.46 (1.87)
Cardio 360 days	2.41 (3.03)	2.56 (3.19)
Adm. 30 days	0.10 (0.30)	0.08 (0.30)
Adm. 90 days	0.24 (0.55)	0.24 (0.56)
Adm. 180 days	0.47 (0.84)	0.46 (0.86)
Adm. 360 days	0.78 (1.19)	0.78 (1.20)
ED 30 days	0.16 (0.43)	0.14 (0.41)
ED 90 days	0.39 (0.76)	0.39 (0.77)
ED 180 days	0.72 (1.19)	0.75 (1.22)
ED 360 days	1.27 (1.82)	1.29 (1.76)
LOS 30 days	8.08 (7.33)	7.54 (5.22)
LOS 90 days	9.70 (8.42)	9.19 (8.05)
LOS 180 days	10.51 (10.32)	11.06 (11.61)
LOS 360 days	12.23 (12.91)	12.58 (13.70)
Hosp. Cost 30 days	14,128 (19,904)	13,612 (12,075)
Hosp. Cost 90 days	15,817 (17,886)	15,224 (14,733)
Hosp. Cost 180 days	17,521 (19,934)	18,762 (20,325)
Hosp. Cost 360 days	21,008 (23,438)	21,768 (22,702)
Observations	987	987
<i>Binary Dep. Variables</i>		
	Intent to Treat	Control
PCP 30 days	0.24 (0.43)	0.21 (0.41)
PCP 90 days	0.41 (0.49)	0.37 (0.48)
PCP 180 days	0.58 (0.49)	0.55 (0.50)
PCP 360 days	0.74 (0.44)	0.73 (0.45)
Cardio 30 days	0.20 (0.40)	0.22 (0.41)
Cardio 90 days	0.42 (0.50)	0.44 (0.50)
Cardio 180 days	0.56 (0.50)	0.61 (0.48)
Cardio 360 days	0.62 (0.49)	0.68 (0.47)
Adm. 30 days	0.09 (0.29)	0.08 (0.46)
Adm. 90 days	0.19 (0.39)	0.19 (0.39)
Adm. 180 days	0.31 (0.46)	0.30 (0.46)
Adm. 360 days	0.45 (0.50)	0.43 (0.50)
ED 30 days	0.13 (0.33)	0.12 (0.32)
ED 90 days	0.27 (0.44)	0.27 (0.45)
ED 180 days	0.42 (0.49)	0.42 (0.50)
ED 360 days	0.57 (0.50)	0.56 (0.50)
	987	987

Notes: Standard Deviation in Parentheses.

^a Indicates that the treatment and control means are statistically different at the 99% level.

of outreach. Our member claims records span the period between July 2013 and March 2016.⁹

We used detailed claims data to generate four continuous variables measuring the cumulative total number of primary care visits, cardiologist visits, number of ED visits, and number of hospitalizations since the date of each wave's outreach.¹⁰ In another set of dependent variables, we analyze the length of stay and the total cost of the hospitalization conditional on hospital admission.¹¹ These six dependent variables are measured at four time windows following outreach initiation – 30 days, 90 days, 180 days, and 360 days. We also generate four binary variables that measure whether an individual had any PCP visit, cardiologist visit, ED visit, and any hospitalization within the respective four time windows after the ITT.

Table 1 provides summary statistics for the dependent and independent variables of interest for our final sample separately for the treatment and control groups. The demographic characteristics are balanced based on T-tests between the groups with the exception of slightly higher risk score – which occurs by design. This rein-

forces that we are observing individuals along the threshold that are very similar in observable characteristics. In terms of dependent variables, we observe that the average number of PCP visits, ED visits, and hospitalizations vary between 0.36–3.42, 0.14–1.29, and 0.08–0.78 among the control group 30–360 days after outreach, respectively.

Fig. 4 displays the fraction of members who visited a primary care physician, a cardiologist, an emergency room or who were hospitalized in the 360 days following outreach. For ease of exposition, the intervention threshold is normalized to zero. Members to the left of the threshold belong to the treatment arm and members to the right of the threshold belong to the control arm. Put differently, the value of the risk score decreases as we move from left to right (from –125 to +125). For each arm, we provide the quadratic fit and combine members in equally sized bins. These graphs provide a visual representation for the discontinuity across treatment and control groups for all outcomes of interest using the raw data. There are a number of insights from this figure. First, emergency department visits and cardiologist visits exhibit a discontinuity, which is sharper for the groups closest to the threshold, and which suggests that the treatment groups use of these outlets is markedly lower even though they are at higher risk. We do not observe any relationship in the case of PCP – even though one of the chief goals of the intervention was to draw members at risk to seek outpatient care (see their primary care physician). Finally, we see some evidence of lower hospitalizations for the treatment group, though there exists significant variation in inpatient stays across the treatment and control group.

⁹ Note that the last two waves (waves 9 and 10) with interventions in May and July of 2015 have fewer than 360 days follow-up period. All waves have a follow-up period of at least 270 days.

¹⁰ Our data does not provide diagnosis codes as part of an ED visit or hospitalization, therefore we are unable to identify the underlying cause of such utilization. Similarly, we are unable to distinguish between new and repeated admissions.

¹¹ We were unable to gain access to total utilization costs, due to privacy concerns from the insurer.

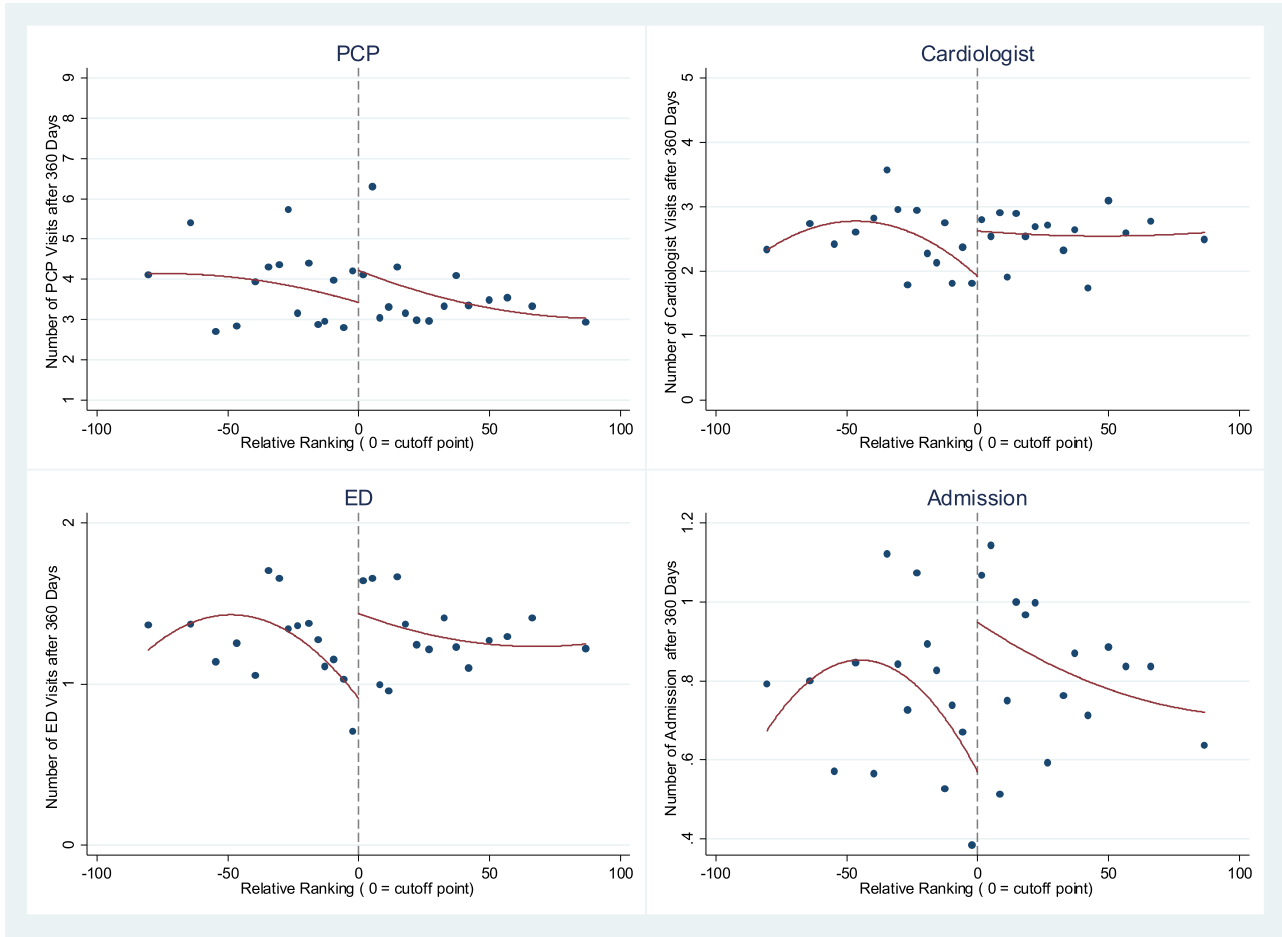


Fig. 4. The Number of Care Utilizations after 360 days of the ITT along the Treatment and Control Samples.

Notes: Fig. 4 displays the unadjusted care utilization patterns of members in the treatment (left of the cutoff) and control group (right of the cutoff) with a quadratic trend.

Methods

We implement a sharp regression discontinuity design to study the impact of selection into outreach on several outcomes. Again, the underlying assumption is that no other variable changes discontinuously at the threshold, and that individuals to the left and right of the cutoff are therefore similar on both observables and unobservable characteristics. Specifically, in our main model we estimate the impact of the outreach using local polynomial regressions with a triangular kernel weight. We limit the sample to a bandwidth of 125 relative to the threshold to capture the causal impact of the outreach among a homogeneous sample. Our regression takes the following form:

$$Y_{it} = \beta_0 + \beta_1 ITT_i + \beta_2 f(R_{it}) + \beta_3 ITT_i \cdot f(R_{it}) + \beta_4 X_{it} + \theta_t + \varepsilon_{it} \quad (1)$$

Where Y_{it} is the outcome for individual i and wave t , measured as member utilization within a 30, 90, 180, or 360 days window post intervention. ITT_i is a dummy variable equal to one if individual i qualified for outreach and zero otherwise. $f(R)_{it}$ is a polynomial function of the relative rank of individual i to the threshold (normalized to zero) at wave t . Our main regression results utilize a quadratic polynomial. $f(R_{it})$ and $ITT_i \cdot f(R_{it})$ allow for different slopes on the treatment and control side of the bandwidth. X_{it} is a vector of control variables that includes all demographics displayed in Table 1 (age, gender, enrolled month, plan choice (HMO vs. PPO), and whether the plan has pharmacy coverage) as well as the risk score for each member i at wave t . θ_t is a wave fixed effect, cap-

turing potential modifications to the algorithms and changes over time. ε_{it} is the error term.¹²

Our model relies on a key assumption to identify the effect of the outreach program. First, the threshold separating treatment and control is chosen independently of the risk score, suggesting it is exogenous and treatment is essentially random. In other words, it needs to satisfy the underlying assumption that no other factors than treatment changes discontinuously at the threshold. Given that the threshold was arbitrary, determined solely by the CM team's resource availability at a given point in time, we do not expect that typical confounders bias our estimates. There is no member level self-selection into the outreach initiation and the threshold is arbitrary (i.e. the care management team choice of capacity was independent of any specific trends in the risk score of eligible individuals across waves).¹³ If this was the case, a discontinuous jump in the risk scores around the cutoff would be visually

¹² The RD design can also be displayed the following way: $Y_{it} = \beta_0 + \beta_1 ITT_i + \beta_2 f(R_{it} - 0) + \beta_3 ITT_i \cdot f(R_{it} - 0) + \beta_4 X_{it} + \theta_t + \varepsilon_{it}$. Also notice that when we evaluate β_1 , the interaction terms of ITT_i with $ITT_i \cdot f(R_{it} - 0)$ disappear because R_i evaluated at the mean is equal to zero. This is the result of the symmetry and balance of the sample in terms of treatment and control units across the threshold.

¹³ We also performed density tests for each wave – resulting in 10 separate tests (see Appendix Fig. 1). In most cases the densities were similar across treatment and control arm. In some cases the density was higher for members with a risk score rank above 100. Overall, there is no clear sign of bunching along the cutoff. The differences in densities across waves can occur because we exclude members who are included in subsequent outreach cohorts and may therefore do not appear again (such approach satisfies that members received the ITT only once).

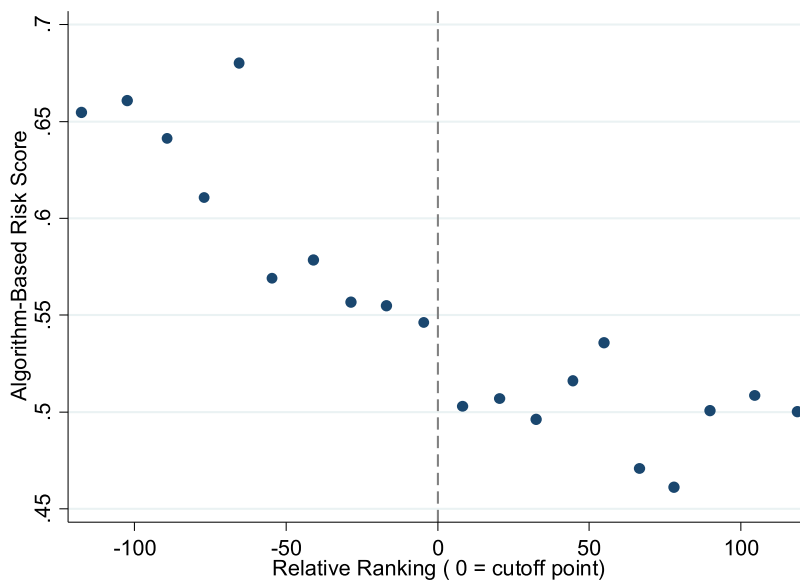


Fig. 5. Risk Score along Treatment and Control Groups.

Notes: Fig. 5 displays the binned risk score for all waves based on the rank order of the risk score. Intent to Treat members are displayed on the left of the cutoff and controls members are displayed on the right.

detected. Fig. 5 refutes this potential confounder and displays the individual risk score relative to the threshold for all waves, where we again centered the threshold at zero. As expected, we observe a negative relationship, as low relative rank (negative numbers, large in absolute value) represent high risk scores, while high relative rank (large numbers) represent lower risk scores. We do not see any discontinuity before or after the threshold along the relative risk ranking and are therefore confident that our RD design passes the initial RD assumption. Performing the analysis separately for each wave, we also do not observe a discontinuous jump around the cutoff. We also estimated a regression of the risk score on a binary treatment variable controlling for the relative rank and do not find a statistically significant jump. As an additional check we also separately estimated regressions in the form presented in Eq. (1) with the control variables in X_i as the dependent variable. In all of these regressions, we do not find any statistically significant impact at the threshold.¹⁴

Second, a potential concern may be that heterogeneity in health coaches' skill level affects the sorting of members into outreach. This would inflate the treatment effect if the more skilled health coaches select to approach the least sick population. This is not possible in our setting, as health coaches were given a list of names, which did not include information on member risk score prior to outreach, thereby mitigating selection concerns about sorting of health coaches to members. Finally, we provide several robustness checks that explore the sensitivity of our results to other potential functional forms.

Results

We separately estimate Eq. (1) for all our dependent variables for all four post ITT windows. Each coefficient estimate in Tables 2 and 3

represents a separate RD regression, totaling 40 regressions for 10 dependent variables.¹⁵

Table 2 presents the impact of outreach initiation on the extensive margin – whether any healthcare service utilization occurred within a certain post outreach window. Each column presents a different dependent variable and each cell represents a different regression. We find no statistically significant impact of outreach initiation on primary care physician visits regardless of post treatment time window. Ex ante, we may have expected increases in PCP utilization given that prompting primary care visits might provide a mechanism for medical intervention that might reduce the need for an ED visit or acute hospital admission. Second, we find large decreases in cardiologist visits across most post-treatment periods. Specifically, we find that treatment reduces cardiologist visits by 14 percentage points within 30 days after outreach and that this effect persists throughout 360 days post treatment. Relative to the 360-day control group average, this drop represents a 20% reduction in cardiologist visits. Third, we find a consistently negative trend, though marginally statistically insignificant reductions in hospital admissions in three out of the four post outreach periods; however, they do represent economically meaningful effects. Given the large cost of admission, even small reductions in admission rates can lead to significant cost savings. Specifically, we find that the outreach program reduces hospital admission by 8.4 percentage points after 90 days, and 3.8 percentage points after 360 days. Fourth, we find large reductions in emergency room use by about 16 percentage points (38%) and 12 percentage points (21%) after 180 and 360 days, respectively.

We now turn to the continuous dependent variable measures in Table 3 that reports the results on the volume of hospitalizations, PCP visits, cardiologist visits, ED visits, length of stay, and hospitalization costs for all four post-outreach initiation windows. Similar to the extensive margin estimates, we only find statistically significant reductions in cardiologist visits and emergency room visits.

¹⁴ We performed local polynomial regressions replacing the dependent variable with one of the five control variables (age, gender, pharmacy coverage, HMO plan, and months enrolled) and did not find statistically significant effects. Specifically, the findings lead to p-values of 0.27 for age, 0.99 for gender, 0.62 for pharmacy, 0.26 for HMO plan, and 0.52 for months enrolled. The coefficients are presented in Appendix Table 17.

¹⁵ We also estimated the impact of ED cost, however, given the relatively small cost per encounter we did not find any impact.

Table 2

Local Polynomial Regression Discontinuity Estimates on the Impact of the Outreach Program on the Probability of Care Utilization within 30, 90, 180, and 360 Days.

	Primary Care Physician	Cardiologist	Hospital Admission	Emergency Room
30 Days	0.010(0.055)	−0.144*** (0.054)	−0.037(0.038)	−0.057(0.042)
90 Days	0.055(0.064)	−0.086(0.064)	−0.084*(0.050)	−0.095(0.058)
180 Days	0.013(0.064)	−0.154*** (0.064)	−0.084(0.063)	−0.157** (0.065)
360 Days	0.015(0.054)	−0.143** (0.062)	−0.038(0.065)	−0.116* (0.066)
N	1974	1974	1974	1974

Note: Each cell entry represents a separate regression including all control variables, for a sample window of 125 left and right of the cutoff. Standard errors in parentheses * indicates $p < 0.1$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$.

Table 3

Local Polynomial Regression Discontinuity Estimates on the Impact of the Outreach Program on the Number of Care Utilizations within 30, 90, 180, and 360 Days.

	Primary Care Physician	Cardiologist	Hospital Admission	Emergency Room	Length of Stay (Conditional on Adm.)	Hospital Cost (Conditional on Adm.)
30 Days	0.178(0.177)	−0.178*** (0.072)	−0.030(0.041)	−0.101* (0.053)	−0.016(2.270)	−5,744(6656)
90 Days	0.324(0.350)	−0.207(0.143)	−0.106(0.066)	−0.207** (0.090)	0.730(1.996)	−3,023(4349)
180 Days	0.286(0.576)	−0.328(0.251)	−0.117(0.108)	−0.356** (0.140)	0.522(2.082)	−1,989(3651)
360 Days	0.387(0.819)	−0.692* (0.404)	−0.156(0.149)	−0.523** (0.213)	−0.126(2.158)	−1,735(3851)
N	1974	1974	1974	1974	864 ^a	864 ^a

^a N = 864 refers to the number of patients with at least one hospitalization with the 360 days following the outreach date. The sample size for 30, 90, and 180 days is 170, 373, and 602, respectively.

The number of ED visits and cardiologist visits decrease by 0.10 and 0.18 visits in the 30 days following outreach initiation, respectively. The rate of ED visits decreases after 360 days by 0.52 visits, which implies a 40% reduction in the average number of ED visits, and the number of cardiologist visits decrease after 360 days by 0.70 visits, suggesting a 27% reduction in the average number of cardiologist visits. Moreover, we find statistically insignificant increases in the number of PCP visits, reductions in the number of hospital admissions, length-of-stay, and hospitalization cost. The economically meaningful but statistically insignificant admissions rate reduction deserves some discussion. The coefficient implies a reduction of admissions for all post-treatment periods from 30 to 360 days. The magnitude suggests a relatively early impact on admissions that persists in the long-run. Setting the estimate in perspective, the impact on the 90-day admission rates is equal to a reduction of 38% for 30 days, 46% for 90 days, 26% for 180 days, and 20% after 360 days relative to the mean admission rate of the control group.¹⁶

Overall, we find large reductions in the probability and the number cardiologist and ED visit utilization that suggest potential substitution away from costly care management services to provider preferred management solutions. While the coefficients on hospital admission are not statistically significant, which may be due to low statistical power, they do present economically meaningful reductions and potential cost savings.

Robustness checks

We conduct several robustness checks to evaluate the sensitivity of our results. We address functional form concerns by estimating local linear regressions and third degree (cubic) local-polynomial regressions. We also estimate regressions that drop the triangular kernel weight, which in essence decreases the contribution of beneficiaries close to the threshold relative to those farther away. We also conduct several robustness checks that increase the sample size, by including individuals farther away from the cutoff, and consider three alternative bandwidth choices – 150, 200, and 250

ranks from the threshold on both sides. Lastly, we estimate regressions utilizing the bandwidth selection procedure by [Imbens and Kalyanaraman, \(2012\)](#), and [Calonico et al. \(2014\)](#). In all cases, the results do not change the conclusions from our main results. We provide estimates for all robustness checks in Appendix Tables 1–14 in Supplementary material.

Heterogeneous effects

Taking advantage of wave-specific regression discontinuity cutoffs, allows us to explore cross-wave treatment heterogeneity. Thus far we have normalized all wave-specific cutoffs to zero and performed a full sample RD analysis on the standardized cutoff using wave fixed-effects. A relatively recent literature has suggested exploring multiple RD cutoffs by representing the dynamics and stability of the RD results across individual waves ([Cattaneo et al., 2016](#); [Bertanha, 2016](#)).

The main regression specification yields a local average treatment effect, which in itself is a weighted average across all wave cutoff points. Hence, there might be a discrepancy between the pooled treatment effect and the average treatment effects for each cutoff. The underlying rationale is that the typical standardized RD approach with wave specific cutoffs obscures variation in treatment heterogeneity among members. To address this, we recover an estimate of the heterogeneous treatment cutoff effect by utilizing the wave-specific fixed effect. Here the underlying assumption is that the wave specific fixed effects picks up all variation that is attributed to the specific wave. More specifically, we use Eq. (1) to parametrize the wave specific treatment effect by combining the overall treatment effect with individual wave-specific fixed effects.¹⁷ The wave-specific fixed effects capture differences in the level impact due to unobservable factors, which is driven by wave specific characteristics and may be due to factors that are correlated with treatment status and risk score cutoff.

We display the wave-specific treatment impacts with standard errors on the probability of having any PCP, cardiologist, hospitalization and ED visit after 360 days in [Fig. 6](#).¹⁸ The upper panel

¹⁶ We also performed analysis on the potential hospitalization savings among the full sample (unconditional on hospitalization), and do not find any statistically significant change in overall hospital cost. Nevertheless, all regressions show a negative sign similar to the results presented in [Table 3](#).

¹⁷ Models that include an interaction of the treatment indicator with the wave fixed effect yield similar results.

¹⁸ Similar trends emerge in the wave-specific treatment effects for the continuous outcome measures.

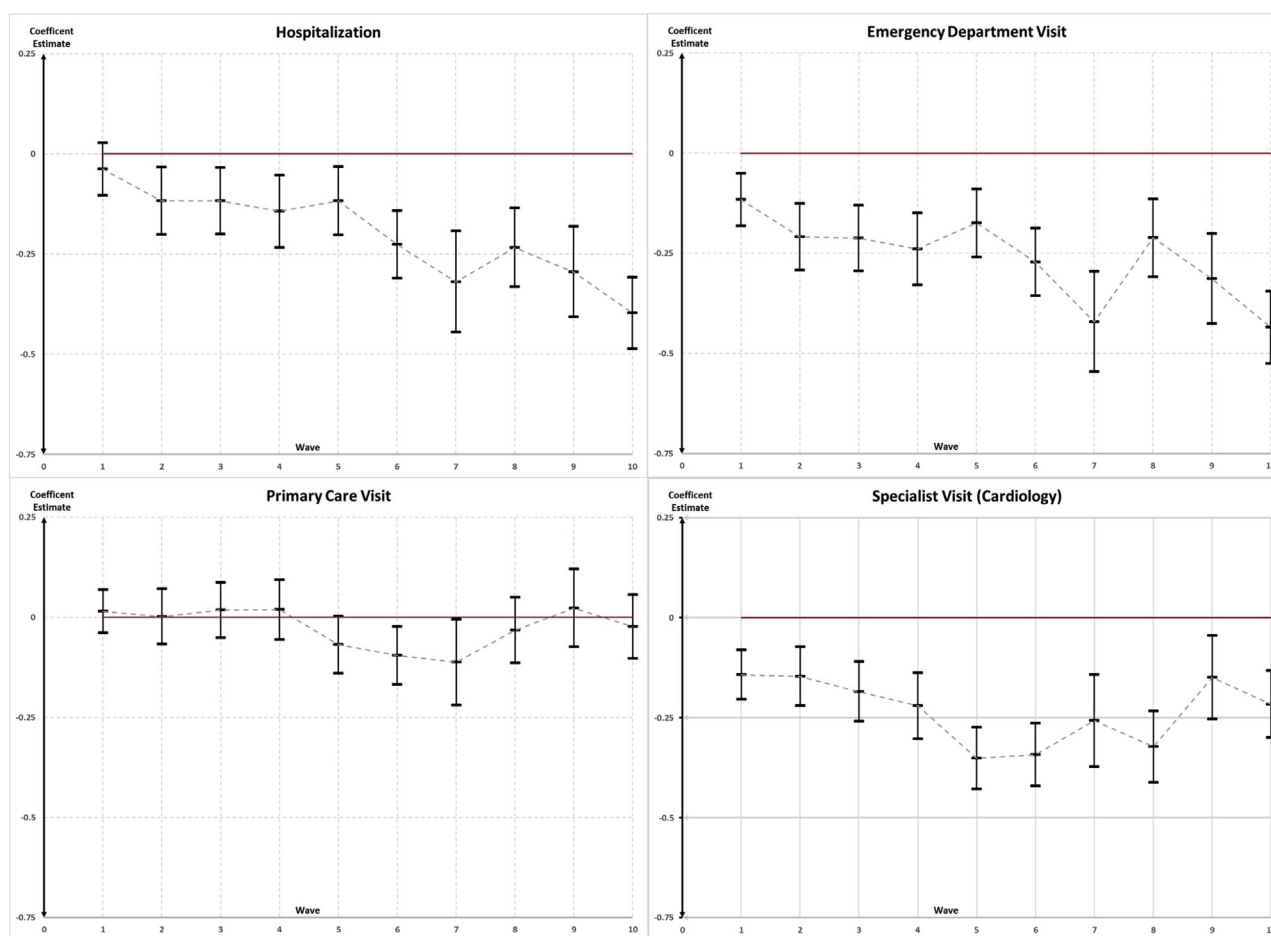


Fig. 6. Wave Specific Estimates for Likelihood of Hospitalizations as well as PCP, Specialist, and ED Visits.

Notes: Fig. 6 displays the wave specific impact of the intent to treat on the probability of having a hospitalization, ED visit, PCP visit, and specialist visit.

Table 4

Local Polynomial Regression Discontinuity Estimates on the Impact of the Outreach Program on the Probability of Care Utilization within 30, 90, 180, and 360 Days (High Risk Score Group).

	Primary Care Physician	Cardiologist	Hospital Admission	Emergency Room
30 Days	0.072(0.086)	−0.111(0.080)	−0.075(0.058)	−0.091(0.066)
90 Days	0.164*(0.096)	−0.105(0.099)	−0.152**(0.074)	−0.190**(0.086)
180 Days	0.005(0.093)	−0.177*(0.096)	−0.188**(0.092)	−0.317*** (0.099)
360 Days	0.038(0.083)	−0.192**(0.093)	−0.109(0.103)	−0.285*** (0.102)
N	995	995	995	995

Note: Each cell entry represents a separate regression including all control variables, for a sample window of 125 left and right of the cutoff. Standard errors in parentheses * indicates $p < 0.1$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$.

displays the wave-specific treatment coefficient estimates for the probability of having a hospitalization and ED visit by wave (x-axis), while the lower panel displays the treatment coefficients for the probability of a PCP and cardiologist visit for waves 1 through 10.

Similar to the overall effect, wave-specific coefficient estimates are negative and statistically significant for cardiologist and ED visits across all waves. However, we also find statistically significant reductions in hospitalizations in waves 6 through 10. The Figure also shows that across all outcomes (less so in the case of cardiology

Table 5

Local Polynomial Regression Discontinuity Estimates on the Impact of the Outreach Program on the Probability of Care Utilization within 30, 90, 180, and 360 Days (Low Risk Score Group).

	Primary Care Physician	Cardiologist	Hospital Admission	Emergency Room
30 Days	−0.076(0.076)	−0.83** (0.080)	0.020(0.057)	−0.043(0.060)
90 Days	−0.023(0.093)	−0.142(0.093)	−0.001(0.075)	−0.008(0.087)
180 Days	−0.049(0.096)	−0.187** (0.093)	0.007(0.088)	−0.046(0.095)
360 Days	−0.081(0.076)	−0.226** (0.090)	0.022(0.095)	−0.042(0.095)
N	979	979	979	979

Note: Each cell entry represents a separate regression including all control variables, for a sample window of 125 left and right of the cutoff. Standard errors in parentheses * indicates $p < 0.1$, ** indicates $p < 0.05$, and *** indicates $p < 0.01$.

gist visits) the wave-specific effects tend to get stronger over time, with later waves having a larger negative coefficient estimate.

We also explore heterogeneity based on whether the risk score cutoff was high or low among waves. Here we estimate stratified samples for waves above the median risk score cutoff of 0.55 (waves 4, 5, 6, 7, 9) and waves below the median risk score cutoff (waves 1, 2, 3, 8 and 10). Tables 4 and 5 display local polynomial estimates for the high and low risk score stratified samples on the probability of care utilization, and Appendix Tables 15 and 16 in Supplementary material display the same regression results for the continuous outcomes. The results clearly show that the high risk score waves saw larger reductions in hospital admission and ED visits compared to the low risk score waves.

The findings of the effectiveness of later waves may be explained in two ways. First, the intervention became more effective over time due to learning on the part of the care management team. Second, the predictive power of the algorithm in identifying individuals that can benefit from the intervention improved over time.¹⁹ The results also imply that the overall treatment effect obscures wave-specific reductions in downstream healthcare utilization that occur predominantly in later waves. Most noticeably in the case of hospitalizations. The reductions in the likelihood of hospitalization in the last five waves translate to a 28% to 50% reduction relative to the control group rate.

Conclusion

Many entities are invested in predictive analytics to elicit insight that can lead to improved quality of care and reductions in cost. Providers and payers increasingly employ risk stratification algorithms to assist in prioritizing clinical workflow, reducing system waste, and managing population health.²⁰ Based on this predictive risk score ranking, care managers are prompted to focus on those members at highest risk and preemptively intervene with medication reconciliation and appointment scheduling.

Providers base their predictive analytics on hospital discharge data and electronic health records, while payers base their predictive analytics on insurance claims data. Provider data are rich in clinical elements important for assessing risk, but are lacking information for medication adherence and healthcare utilization in other outlets, which is at the heart of claims data. Put differently, while recognizing the potential to realize efficiencies through predictive analytics, both payers and providers use incomplete data to direct potentially timely interventions.

This paper utilizes proprietary claims data from high risk Medicare Advantage members diagnosed with congestive heart failure to examine the effect of a proactive outreach program on hospitalizations, emergency room use, primary care and cardiologist visits. Using a Regression Discontinuity strategy that exploits the quasi-random selection into outreach initiation we estimate the impact of outreach on healthcare utilization. Our results suggest that the outreach program reduced cardiologist visits, and emergency room visits along all post-intervention periods. The magnitude of the estimates suggest that the outreach program was responsible for a 20% reduction in the likelihood of a cardiologist visit, and a 21% reduction in the likelihood of an emergency room visit over the first 360 days following outreach. Similarly, we find across the board reductions in the number of cardiologist visits and the number of emergency room visits. We also find wave-specific reductions in the likelihood of hospitalization, which is concentrated in later waves,

potentially stemming from improvement in algorithm capabilities and learning by care management teams. We also observe that the effects were concentrated among waves with higher than average risk scores, suggesting that the impact of the intervention has been especially effective among people who are at higher risk of care needs. In other words, the effectiveness of the care management program may also rely on a member population with at least a moderate risk score, which allows the care management team to close gaps in care that can materially affect the utilization patterns of the member. These findings suggest that continuity of care, in the form of a communication hub as part of a targeted intervention, may serve as an effective tool in reducing unnecessary utilization.

While we find evidence that efforts to close gaps in care can reduce potentially avoidable ED visits among chronically ill elderly members, this effect is not manifested through increased reliance on primary care, as the number and likelihood of primary care visits is not affected by outreach. Our findings may be driven by improved care coordination, which led patient visits to be concentrated among fewer providers. Reduced primary care fragmentation, for example, may serve as a mechanism translating improved care coordination into reductions in hospital and emergency room utilization. It is also possible that the intensity of treatment, and patient involvement may change as a result of the intervention and may explain our findings.

Our results fit well within the range of evidence on successful case management interventions. Brown et al. (2012) display that effective case management programs have six distinguishing features relative to ineffective programs. Five out of the six features were present in the intervention covered in this paper - care manager serve as a communication hub, evidence based educational intervention, comprehensive medication management, timely transition of information of changes in care settings of members, and care coordination based on members diagnoses during hospitalizations. Including these aspects in a care management have shown to reduce hospitalization among Medicare members between 8%–15% (Brown et al., 2012). As a result, this study has two major implications for future policies that try to improve patient health and reduce potentially unnecessary care. First, targeted interventions identifying high risk members with potentially costly inpatient care needs can display large gains in terms of reductions of potentially unnecessary care. Second, effective telehealth case management, rather than face to face case management, can lead to reductions in ED care.

One potential mechanism driving the reduction in ED visits may be the result of an increased outpatient provider concentration. We are unable to decompose whether the number of outpatient visits shifted towards one provider while the total number of visits remains constant. Several studies present evidence highlighting the importance of care management in terms of frequency and provider concentration. Another potential explanation may be that care managers' coordinated approach to care rendered additional primary care visits unnecessary. Many of the members' questions can be answered effectively, relying on care managers' broad educational and clinical knowledge.

It remains an open question why the intervention did not largely impact hospitalizations in early waves given that it was an explicit target of the intervention. An explanation for the null result may be that initially the algorithm and care management team did not effectively identify and treat those members at risk for avoidable hospitalizations (McWilliams and Schwartz, 2017). Thus, these members did not seem to respond to the intervention or close gaps of care in a manner that reduces the costliest healthcare encounters. An important next step in proactive outreach programs requires that those most likely with the greatest need of care be clearly identified and persuaded to close gaps of care. The current program does not seem to deliver on that margin.

¹⁹ The algorithm was revised four times across the ten waves, with additional data components introduced in later versions to provide better predictions.

²⁰ In addition to risk scores stratification, predictive analytics may center around scenario simulations and geo-spatial analysis.

Overall, the intervention can be seen as a partial success for the provider. Most ED care, especially among members with manageable conditions, has been widely recognized as unnecessary and expensive care. However, it is not clear whether the reductions in care utilization has been appropriately replaced with care management outside the ED or made unnecessary through better care coordination. Both potential mechanisms imply cost savings (replacing ED with outpatient care and a reduction in care), however it is also possible that ED care is not properly replaced with suitable primary care. ED visits may display especially needed care for members that would have otherwise gone undetected. Similarly, we are also only able to infer the short-term impact of the intervention on utilization and indirectly patient health. It is conceivable that the care organization efforts may lead to long-term adverse health effect— if avoidable ED care was not properly replaced with care in other outpatient settings.

Lastly, from a provider's welfare perspective our main results also imply savings 30 days post-outreach of \$500 to \$1000 for emergency department visits per member per year. Similarly, while mostly statistically insignificant, the coefficient estimates for reduction in hospitalization implies annual savings of approximately \$13,000 per member over the thirty days post-outreach period. These generally large effects may be the result of targeting the sickest members within a high spending population, who based on these predictive models have the most to gain from an intervention.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhealeco.2019.02.002>.

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